EXP-02

7/26

Study of Different Types of Network Cables.

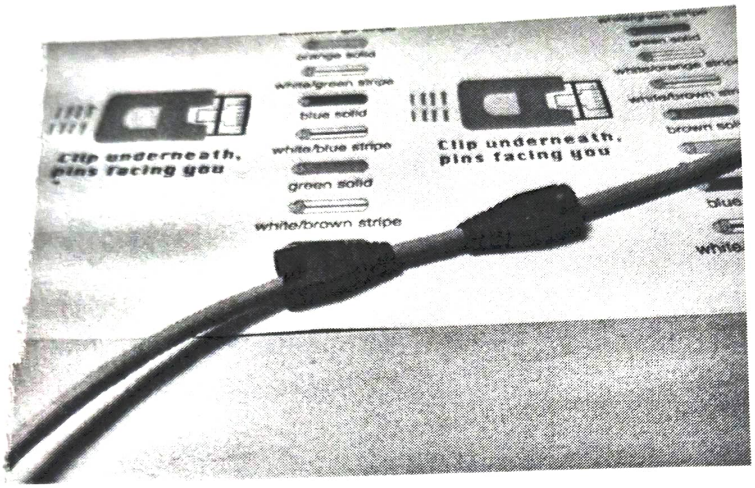
AIM:

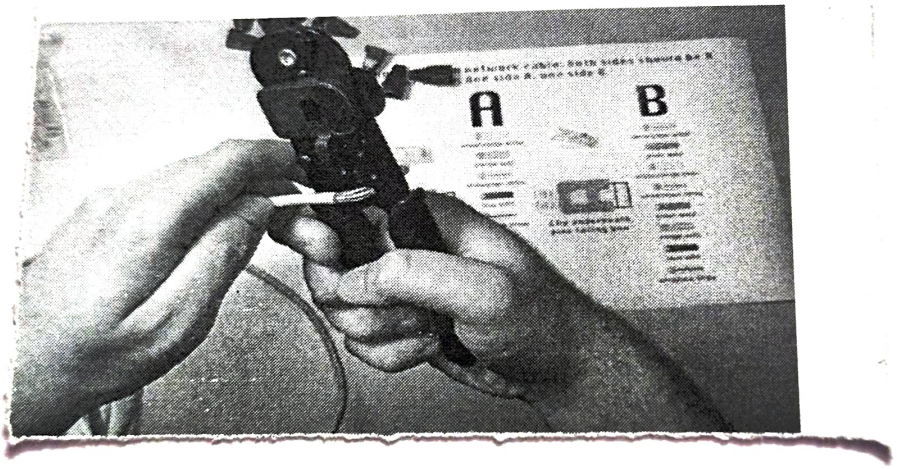
To study and understand different types of network cables, TO make your own ethernet cross over Cable /Stright Cable

a) Understand different types of Network Cable Different types of Cables used in networking are

1. Unshielded Twisted Pairs (UTP) Cables.
2. Shielded Twisted Pairs (STP) Cables
3. Coaxial Cable.
4. Fiber optic Cable.

CABLE	CATEGORY	maximum Data Transmision	ADVANTAGES / DISADVANTAGE	APPLICATION
UTP	Category 3	10 bps	Advantages * Cheaper in cost * Easy to install as they have a smaller overall diameter Disadvantages more prone to EMI and noise	10 Base-T Ethernet
	Category 5	Up to 100 MBPS		Fast ethernet
	Category 5e	1 Gbps		Gigabit ethernet.
STP	Category 6, 6a	10 Gbps	ADVANTAGES * Shielded * Faster than UTP less susceptible to noise and interference DisAdvantage * Expensive * Greater installation effort	Gigabit ethernet Low Ethernet (55m) widely used in data Center
SSTP	Category	10 Gbps.		Gigabit Ethernet (100m)


Clip underneath, pins facing you
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
Clip underneath, pins facing you


A
B

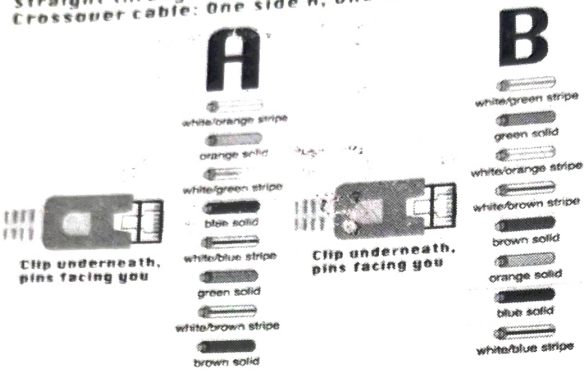
Coaxial cable	RG-6 RG-59 RG-11	10 - 100 Mbps	* High bandwidth * Immune to interference * Low cost bandwidth * Versatile DISADVANTAGE * Limited distance * Cost * Size is bulky	Speed of Signal is Television network High speed Internet Connection
Fibre optics Cable	Single mode Multimode	100Gbps	ADVANTAGES * High Speed * High bandwidth * High Security * Long distance DISADVANTAGES * Expensive * Required Skilled Installers	Maximum distance of fibre optics cable is around 100 meters

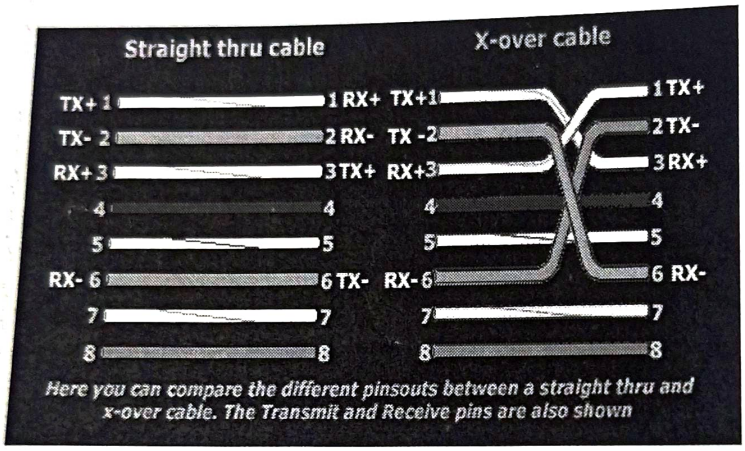
Make Your Own Ethernet Cross-over CABLE / STRAIGHT @ CA

TOOLS AND Parts Needed :-

* Ethernet Cabling - CAT5E CAT5E is certified for Giga bi Support, but CAT5 cabling works as well, just over shorter distance.
* Crimping tool, This is an in-one network tool shaped to push down the pin in the plug and clip And cut the shielding off the cables
* Two RJ45 Plugs
* Optional two plug shields

Straight through network cable: both sides should be A
Crossover cable: One side A, one side B





Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

STEPS:

1. To start constructions of the device, begin by threading shields onto the cable.

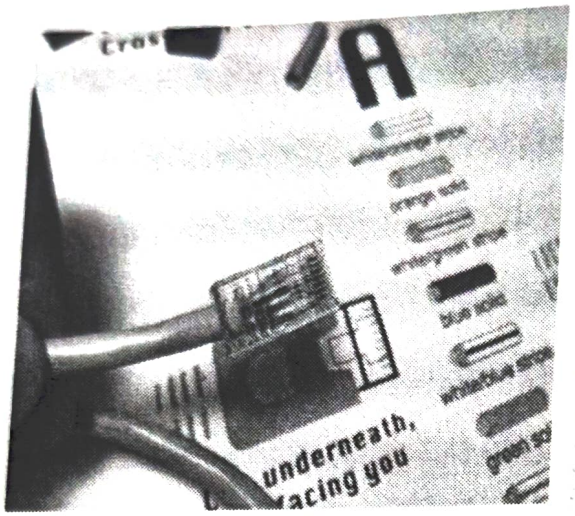
2, Next, strip approximately 1.5cm of cable shielding from both ends, The crimping tool has a round area to complete the task

3 After, you will need to untangle the wires, there should be four twisted pairs. Referring back to the sheet arrange them from top to bottom, one end should be in arrangement A and the other in B

4. Strengthen the wires from there evenly and keep the order correct, Hold the RJ45 plug with clip away and gold pins facing you slide wires in carefully without disturbing the sequence.

5. Next put the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't that meant that you stripped off two much shielding simply snip the cable back a little more

6. After the wires are securely sitting inside the plug, insert it into the crimping tool and push down. It should sharp correctly but pushing too hard, can crack the fragile plastic plug.


underneath,
facing you

Step 7: Lastly repeat for the other end using diagram B (To make a crossover) cables / using diagram A (to make a straight through cable)
to test it plug it in an attempt to connect two devices directly.

RESULT:

Thus different types network cables were studied successfully and Straight through and Cross over Ethernet cables were prepared and test successfully.